

Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Uses $A = \frac{1}{2}ab\sin C$ for triangle OAC or OAB PI by equation	AO1.2	B1	$\frac{1}{2}r \times \frac{r}{2} \sin \theta = \frac{1}{4} \left(\frac{1}{2}r^2\theta \right)$ $\Rightarrow \frac{r^4}{4} \sin \theta = \frac{1}{8}r^2\theta$ $\Rightarrow 2r^2 \sin \theta = r^2\theta$ $\Rightarrow 2 \sin \theta = \theta$ AG
	Forms an equation relating the area of OAC and ABC in the form $Ar^2 \sin \theta = Br^2\theta$	AO3.1a	M1	
	Obtains fully correct equation ACF	AO1.1b	A1	
	Simplifies to obtain required equation, only award if all working correct with rigorous argument.	AO2.1	R1	
(b)	Rearranges to the form $f(\theta) = 0$ PI by correct θ_2 or θ_3	AO1.1a	M1	$f(\theta) = \theta - 2 \sin \theta = 0$ $\theta_{n+1} = \theta_n - \frac{\theta_n - 2 \sin \theta_n}{1 - 2 \cos \theta_n}$ $\theta_2 = 2.094395\dots$ $\theta_3 = 1.913222\dots$ $\theta_3 = 1.91322 \text{ (5 d.p.)}$
	Differentiates their $f(\theta)$ or uses calculator PI correct θ_2 or θ_3	AO1.1b	A1	
	Obtains correct θ_3	AO1.1b	A1	
(c)	Obtains percentage error for θ_3 AWRT 0.94%	AO3.2b	B1	0.935%
				Total 8 marks

Q2.

Marking Instructions	AO	Marks	Typical Solution
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(a) (i)	Uses model to form equation with $V = 0$	AO3.4	M1	$\therefore 10 + 100\left(\frac{T}{30}\right)^3 - 50\left(\frac{T}{30}\right)^4 = 0$
	Rearranges to isolate T^4 term	AO1.1a	M1	$\Rightarrow 50\left(\frac{T}{30}\right)^4 = 10 + 100\left(\frac{T}{30}\right)^3$
	Completes rigorous and convincing argument to clearly show the required result. Need to see evidence of division by T to isolate T^3 term Must be an equation throughout AG	AO2.1	R1	$\Rightarrow \frac{T^4}{16200} = 10 + \frac{T^3}{270}$ $\Rightarrow \frac{T^3}{16200} = \frac{10}{T} + \frac{T^2}{270}$ $\Rightarrow T = \sqrt[3]{\frac{162000}{T} + 60T^2}$
(ii)	Calculates T_1 (44.96345.....)	AO1.1a	M1	$T_1 = 44.963$ $T_2 = 49.987$
	Calculates T_2 and T_3 (49.98742....) Condone greater than 3dp (53.50407....)	AO1.1b	A1	$T_3 = 53.504$
(iii)	Explains 38 in context	AO3.2a	B1	38 represents current year 2018

(b)	Translates 2029 into $t = 49$	AO3.3	B1	$10 + 100\left(\frac{t}{30}\right)^3 - 50\left(\frac{t}{30}\right)^4 = 4.5 \times 1.063$
	Uses models to set up equation or evaluate both models at one value of t	AO3.4	M1	$\Rightarrow t = 49.009$ $1980 + 49 = 2029$
	Obtains correct values for both models for two appropriate values of t . $t \in [49, 50]$ eg $t = 49$ and $t = 50$ $t = 49$ gives: 89.89 and 89.81 $t = 50$ gives: 87.16 and 95.47 Or Solves equation using any method to obtain AWWF 49.009 to 49.01	AO1.1b	A1	Therefore use of oil and production of oil will be equal in the year 2029
	Explains that the use of oil and the production of oil are equal when $t = 49.009$ Or	AO2.4	E1	

Uses a change of sign argument OE to explain that the value of each model for two appropriate values of t shows that the production of oil and the use of oil are the same for $t \in (49,50)$			
Total 10 marks			

Q3.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Starts an argument by showing that $f(-2) < 0$ and $f(-1) > 0$ Both attempted and at least one evaluated correctly f must be clearly defined or substitution of values must be explicit.	AO2.1	R1	$f(x) = x^3 - 3x + 1$ $f(-2) = (-2)^3 - 3(-2) + 1 = -1 < 0$ $f(-1) = (-1)^3 - 3(-1) + 1 = 3 > 0$ Change of sign and $f(x)$ is continuous so a root must lie between $x = -2$ and $x = -1$
	Explains reasoning fully to complete the argument Evaluations above need to be of opposite sign and 'change of sign' OE seen and reference to x -values -2 & -1 and reference to continuous function	AO2.4	E1	
(b)	Uses Newton–Raphson, must have $f'(x)$ correct PI by correct substitution	AO1.1a	M1	$x_{n+1} = x_n - \frac{(x_n)^3 - 3x_n + 1}{3(x_n)^2 - 3}$ $x_2 = -2 - \frac{-1}{3(-2)^2 - 3}$ $= -\frac{17}{9}$
	Substitutes $x_1 = -2$ into 'their' Newton–Raphson formula (accept 'their' $f(-2)$ from part (a))	AO1.1a	M1	
	Obtains correct value for x_2 $-\frac{17}{9}$ or $-1\frac{8}{9}$ or -1.89 or better	AO1.1b	A1	

(c) Explains why the method fails when $x_1 = -1$ This must include a substitution of $x_1 = -1$ and an explanation of what goes wrong eg division by zero not possible division by zero method fails	AO2.4	E1	$x_1 = -1$ $x_2 = -2 - \frac{3}{3(-1)^2 - 3} = -2 - \frac{3}{0}$ causes division by zero (expression undefined) ALT $f'(-1) = 0$, function has zero gradient at this point, method will fail
Total 6 marks			

Q4.

(a) $f(2) = -3$

$f(3) = 10$

*$f(x) = x^3 - 6x + 1$ must have both values correct
allow $f(2) < 0$ and $f(3) > 0$
only if $f(x)$ is defined and no errors seen*

M1

change of sign $\Rightarrow 2 < \alpha < 3$

*must have **both statement and interval**
which may be written in words / symbols*

A1

(b) $x^3 = 6x - 1$

or $x^2 - 6 + \frac{1}{x} = 0$

or $x^2 - 6 = -\frac{1}{x}$

must see one of these lines and no errors

$x^2 = 6 - \frac{1}{x}$

AG

B1

1

(c) $x_2 = \sqrt{6 - \frac{1}{2.5}} = 2.366(432)$

at least 4sf needed
PI by correct x_3

B1

$$x_3 = 2.362$$

SC1 if B0B0 scored and $x_3 = 2.3617$

B1

2

[5]

Q5.

(a)

$$\left. \begin{array}{l} f(3) = \\ -0.2(18) \\ f(4) = \\ 0.01(83) \end{array} \right\}$$

$$f(x) = e^{-x} - 2 + \sqrt{x}$$

Both values correct

M1

Change of sign $\Rightarrow 3 < \alpha < 4$

Must have **both statement and interval** in words or symbols.

A1

2

(b) $(x_{n+1} = (2 - e^{-x_n})^2 \quad x_1 = 3.5)$

$$(x_2 = 3.8801\dots)$$

$$x_2 = 3.880$$

Do **not** accept 3.88

B1

$$(x_3 = 3.9178\dots)$$

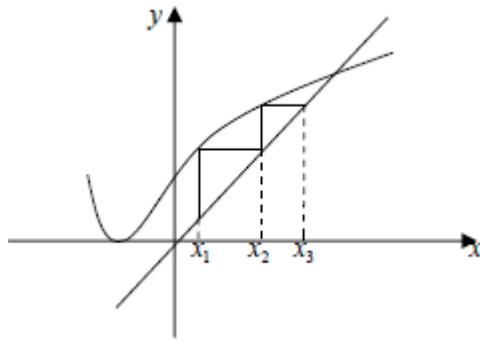
$$x_3 = 3.918$$

Do **not** accept 3.917

B1

2

(c)



Staircase to curve from x_1 including at least two stairs between curve and line $y = x$.

M1

x_2 and x_3 marked on the x -axis.

Do **not** accept marking on the curve or on the line.

A1

2

[6]

Q6.

$$(x_2 =) 10 - \frac{(10^3 - 10^2 + 4 \times 10 - 900)}{(3 \times 10^2 - 2 \times 10 + 4)}$$

$10 - \frac{f(10)}{f'(10)}$ with a correct numerical expression or value
PI for $f(10)$.

B1

$$\begin{aligned} &= 10 - \frac{1000 - 100 + 40 - 900}{300 - 20 + 4} \\ &= 10 - \frac{40}{284} = 10 - 0.1408... \end{aligned}$$

$10 - \frac{f(10)}{f'(10)}$ with a correct numerical expression or value
PI for $f'(10)$.

B1

(= 9.85915....) = 9.859 (to 4 sf)
Must be 9.859

B1

[3]

Q7.

- (a) Tangent at A drawn

At least as far as meeting the x -axis.

Accept reasonable attempt. Must not be linked to P .

B1

1

(b) $x_0 = 2 - \frac{-10}{8}$

PI by 3.25 or $26/8$ OE

M1

$\dots = 3.25$

CAO Must be 3.25

A1

2

[3]

Q8.

(a) $f(x) = 4 \ln x - \sqrt{x}$

Or reverse

$$\left. \begin{array}{l} f(0.5) = -3.5 \\ f(1.5) = 0.4 \end{array} \right\} \text{ must have both values correct}$$

Allow $f(0.5) < 0$ and $f(1.5) > 0$ only if $f(x)$ defined

M1

Change of sign $\therefore 0.5 < \alpha < 1.5$

$f(x)$ must be defined and all working correct, including both statement and interval (either may be written in words or symbols)

OR comparing 2 sides:

$$\left. \begin{array}{l} 4 \ln 0.5 = -2.8 \quad \sqrt{0.5} = 0.7 \\ 4 \ln 1.5 = 1.6 \quad \sqrt{1.5} = 1.2 \end{array} \right\} \text{ (M1)}$$

At 0.5, $LHS < RHS$; at 1.5, $LHS > RHS$

$\therefore 0.5 < \alpha < 1.5$ (A1)

A1

2

(b) $\ln x = \frac{\sqrt{x}}{4}$ or $x^4 = e^{\sqrt{x}}$

Must be seen

$$x = e^{\frac{\sqrt{x}}{4}}$$

AG; no errors seen

B1

1

(c) $x_2 = 1.193$

B1

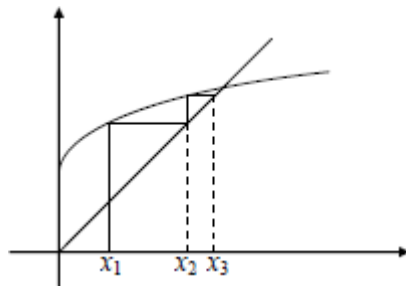
$$x_3 = 1.314$$

If B0B0 scored but either value seen correct to 2 or 4 dp, score SC1

B1

2

(d)



Vertical line from x_1 to curve (condone omission from x -axis to $y = x$) and then horizontal to $y = x$

M1

2nd vertical and horizontal lines, and x_2, x_3 (not the values) must be labelled on x -axis

A1

2

[7]

Q9.

(a) Let $f(x) = 24x^3 + 36x^2 + 18x - 5$

$$f(0.1) = -2.816, f(0.2) = 0.232$$

Both attempted and at least one evaluated correctly to at least 1sf rounded or truncated OE fraction

M1

Change of sign so α lies between 0.1 and 0.2

Need both evaluations correct to above degree of

accuracy and 'change of sign OE' and relevant reference to 0.1 and 0.2

A1

2

(b) $f'(x) = 72x^2 + 72x + 18$
PI

B1

$(x_2=)$

$$0.2 - \frac{24(0.2)^3 + 36(0.2)^2 + 18(0.2) - 5}{72(0.2)^2 + 72(0.2) + 18}$$

B1 for numerator in correct formula

B1

B1 for denominator in correct formula

B1

$= 0.1934$ (to 4dp)

CAO Must be 0.1934

Do not apply ISW

NMS scores 0 / 4

B1

4

[6]

Q10.

(a) $f(x) = (x^2 - 4) \ln(x + 2) - 15$

Or reverse

$$\left. \begin{array}{l} f(3.5) = -0.9 \\ f(3.6) = 0.4 \end{array} \right\}$$

$$\left. \begin{array}{l} f(3.5) = 0.9 \\ f(3.6) = -0.4 \end{array} \right\} \quad M1$$

M1

Attempt at evaluating both $f(3.5)$ and $f(3.6)$

But must see $f(x) = 15 - (x^2 - 4) \ln(x + 2)$

before A1 may be earned

Condone

$$\left. \begin{array}{l} f(3.5) < 0 \\ f(3.6) > 0 \end{array} \right\} \text{Only if } f(x) \text{ defined} \quad M1$$

Or

$$\left. \begin{array}{l} x = 3.5 \quad y = 14.1 \quad (< 15) \\ x = 3.6 \quad y = 15.4 \quad (> 15) \end{array} \right\} \quad M1$$

Change of sign, $\therefore 3.5 < \alpha < 3.6$ OE

Either side of 15, $\therefore 3.5 < \alpha < 3.6$ OE A1

A1

2

(b) $(x^2 - 4) \ln(x + 2) = 15$

$$x^2 - 4 = \frac{15}{\ln(x + 2)}$$

$$x^2 = 4 + \frac{15}{\ln(x + 2)}$$

$\left. \begin{array}{l} \text{Either of these lines correct} \\ \text{Condone poor use of brackets} \\ \text{for M1 only} \end{array} \right\}$

M1

$$x = \pm \sqrt{4 + \frac{15}{\ln(x + 2)}}$$

AG

Must have **both** middle lines and no errors seen

A1

2

(c) $(x_1 = 3.5)$

$$x_2 = 3.578 \quad \text{CAO}$$

B1

$$x_3 = 3.568 \quad \text{CAO}$$

B1

2

Sight of AWRT 3.58 or 3.57 scores B1 B0

Or ± 3.578 or ± 3.568 scores B1 B0

$x_1 = 3.578, x_2 = 3.568$ scores B1 B0

Q11.

(a) $f(x) = \cos^{-1}(2x - 1) - e^x$
or reverse

$$\left. \begin{array}{l} f(0.4) = 0.3 \\ f(0.5) = -0.1 \end{array} \right\}$$

sight of ± 0.3 (AWRT) AND ∓ 0.1 (AWRT)

M1

change of sign $\therefore 0.4 < \alpha < 0.5$

CSO, note $f(x)$ must be defined,
condone $0.4 \leq \alpha \leq 0.5$

A1

2

alternative method

$$\left. \begin{array}{l} e^{0.4} = 1.5, \cos^{-1}(2 \times 0.4 - 1) = 1.8 \\ e^{0.5} = 1.65, \cos^{-1}(2 \times 0.5 - 1) = 1.57 \end{array} \right\}$$

(M1)

$$\left. \begin{array}{l} \text{at } 0.4 \ e^x < \cos^{-1}(2x - 1) \\ \text{at } 0.5 \ e^x > \cos^{-1}(2x - 1) \end{array} \right\}$$

$$\therefore 0.4 < \alpha < 0.5$$

(A1)

(b) $\cos^{-1}(2x - 1) = e^x$

$$2x - 1 = \cos(e^x)$$

$$x = \frac{1}{2}(\cos(e^x) + 1) = \frac{1}{2} + \frac{1}{2}\cos(e^x)$$

AG

must see middle line, and no errors seen,
but condone $\cos e^x$

B1

1

(c) $x_1 = 0.4$

$$x_2 = 0.539$$

CAO

B1

$$x_2 = 0.428$$

CAO

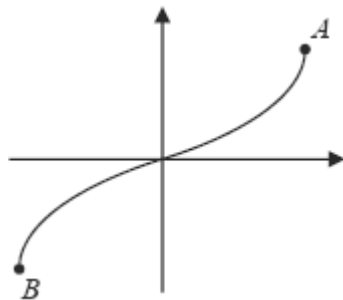
B1

2

[5]

Q12.

(a) (i)



*correct shape passing through origin
and stopping at A and B*

B1

$$A\left(1, \frac{\pi}{2}\right)$$

B1

$$B\left(-1, -\frac{\pi}{2}\right)$$

SC $A(1, 90)$ and $B(-1, -90)$ scores B1

B1

3

- (ii) line intersecting their curve (positive gradient, positive y intercept)

M1

Correct statement

*one solution only, stated or indicated on
sketch - must be in the first quadrant
(ie curve intersects line once)
Must have scored B1 for graph in (a)(i)*

A1

(b)
$$\left. \begin{array}{l} \text{LHS}(0.5) = 0.5 \quad \text{RHS}(0.5) = 1.1 \\ \text{LHS}(1) = 1.6 \quad \text{RHS}(1) = 1.3 \end{array} \right\}$$

M1

At 0.5 LHS < RHS, At 1 LHS > RHS

$\therefore 0.5 < \alpha < 1$

CSO

A1

or

$$f(x) = \sin^{-1}(x) - \frac{1}{4}x - 1$$

f(x) must be defined

$$\left. \begin{array}{l} f(0.5) = -0.6 \\ f(1) = 0.3 \end{array} \right\} \text{AWRT}$$

Allow $f(0.5) < 0$ $f(1) > 0$

(M1)

Change of sign $\Rightarrow 0.5 < \alpha < 1$

(A1)

or

$$f(x) = \sin\left(\frac{1}{4}x + 1\right) - x$$

f(x) must be defined

$$\left. \begin{array}{l} f(0.5) = 0.4 \\ f(1) = -0.1 \end{array} \right\} \text{Attempt}$$

(M1)

Change of sign $\Rightarrow 0.5 < \alpha < 1$

(A1)

or

$$f(x) = 4 \sin^{-1} x - x - 4$$

f(x) must be defined

$$\left. \begin{array}{l} f(0.5) = -2.4 \\ f(1) = 1.3 \end{array} \right\} \text{Attempt}$$

(M1)

Change of sign $\Rightarrow 0.5 < \alpha < 1$

(A1)

2

(c) (i) $x_2 = 0.902$

Sight of AWRT 0.902 or AWRT 0.941

M1

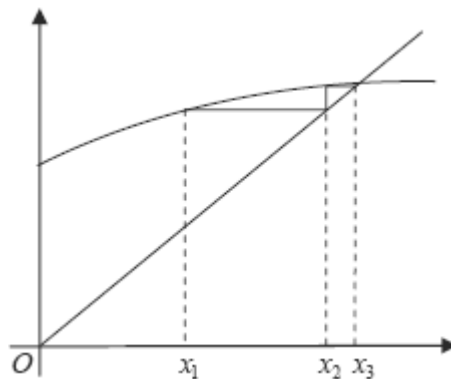
$x_3 = 0.941$

These values only

A1

2

(ii)



*Staircase, (vertical line) from x_1 to curve,
horizontal to line, vertical to curve*

M1

x_2, x_3 approx correct position on x -axis

A1

2

[11]

Q13.

(a) $f(x) = 3^x - 10 + x^3$ (or reverse)

$f(1) = -6$

$f(2) = 7$

Attempt to evaluate $f(1)$ and $f(2)$

M1

Change of sign $\therefore 1 < \alpha < 2$

All working must be correct plus statement

A1

OR

LHS (1) = 3 RHS (1) = 9

LHS (2) = 9 RHS (2) = 2

Must be these values

(M1)

At 1 LHS < RHS, At 2 LHS > RHS

$\therefore 1 < \alpha < 2$

(A1)

2

(b) (i) $3^x = 10 - x^3$
 $x^3 = 10 - 3^x$

This line must be seen

$x = \sqrt[3]{10 - 3^x}$

AG

B1

1

(ii) $(x_1 = 1)$
 $x_2 = 1.913$

Sight of AWRT 1.9 or AWRT 1.2

M1

$x_3 = 1.221$

Both values correct

A1

2

[5]

Q14.

(a) $f(x) = x^3 + 5x - 4$
 $f(0.5) = -1.375$

Condone $f(0.5)$ rounding to -1.4

M1

$f(1) = 2$

Change of sign $\therefore 0.5 < \alpha < 1$

Both statements needed

A1

2

(b) $x^3 + 5x - 4 = 0$

$5x = 4 - x^3$

Must be seen

$x = \frac{1}{5}(4 - x^3)$

AG

B1

1

(c) $x_1 = 0.5$

$(x_2 = 0.775) \left(= \frac{31}{40} \right)$

For x_2 or $x_3 = (2 \text{ sf})$

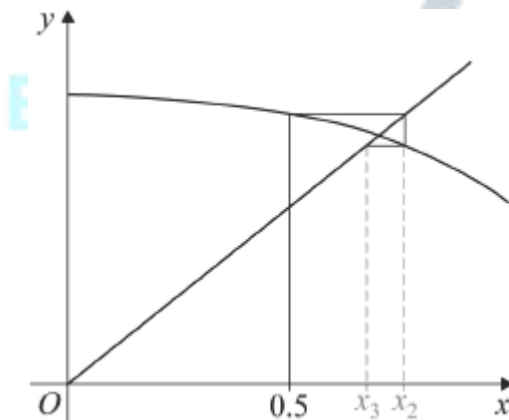
M1

$x_3 = 0.707$

A1

2

(d)



From 0.5 vertical to curve then horizontal to line

M1

CAO

A1

2

[7]

Q15.

(a) (i) $f(x) = \frac{\cos x}{2x+1} - \frac{1}{2}$

OE

$$f(0) = \frac{1}{2}; f\left(\frac{\pi}{2}\right) = -\frac{1}{2}$$

$$x=0 \quad LHS=1, \quad x=\frac{\pi}{2} \quad LHS=0$$

M1

Change of sign $0 < \alpha < \frac{\pi}{2}$

Either side of $\frac{1}{2}, \therefore 0 < \alpha < \frac{\pi}{2}$

A1

2

(ii) $\frac{\cos x}{2x+1} = \frac{1}{2}$
 $\left. \begin{array}{l} 2\cos x = 2x+1 \\ 2\cos x - 1 = 2x \end{array} \right\} \text{ or, } \cos x = x + \frac{1}{2}$

Either line

$$x = \cos x - \frac{1}{2}$$

$$\cos x - \frac{1}{2} = x$$

AG; or

All correct with no errors

B1

1

(iii) $x_1 = 0$
 $x_2 = 0.5$

Attempt at iteration

(allow $x_2 = -0.5, x_3 = 0.38, 0.4$)

M1

$$x_3 = 0.378$$

CAO

A1

2

(b) (i) $\frac{dy}{dx} = \frac{(2x+1)(-\sin x) - \cos x \times 2}{(2x+1)^2}$

Attempt at quotient rule:

$$\frac{\pm (2x+1)\sin x \pm 2\cos x}{(2x+1)^2}$$

M1

Either term correct

A1

All correct ISW

A1

3

(ii) $x = 0$
 $\frac{dy}{dx} = -2$

Correctly subst. $x = 0$ into their $\frac{dy}{dx}$

m1

$\therefore$ Gradient of normal = $\frac{1}{2}$
 CSO

A1

2

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[10]

Q16.

(a) $x + (1+3x)^{\frac{1}{4}} = 0$

$f(-0.32) = 0.1$

$f(-0.33) = -0.01$

AWRT; allow +ve, -ve

M1

Change of sign $\therefore -0.33 < x < -0.32$

A1

2

(b) $x = -(1 + 3x)^{\frac{1}{4}}$
 $x^4 = 1 + 3x$
Attempt to isolate x^4

M1

$$\frac{x^4 - 1}{3} = x$$

AG

A1

2

(c) $x_1 = -0.3$
 $(x_2 = -0.331)$ AWRT

M1

$(x_3 = -0.329)$ AWRT

A1

$x_4 = -0.329$

A1

3

[7]

Q17.

(a) (i) $(-1 + h)^3 = -1 + 3h - 3h^2 + h^3$
PI

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$y_B = (-1 + 3h - 3h^2 + h^3) + 1 - h + 1$
ft numerical error

B1F

$\dots = 1 + 2h - 3h^2 + h^3$
convincingly shown (AG)

B1

3

(ii) Subtraction of 1 and division by h

M1M1

Gradient of chord = $2 - 3h + h^2$

A1

3

(iii) As $h \rightarrow 0$, $\text{gr}(\text{chord}) \rightarrow \text{gr}(\text{tgt}) = 2$

E0 if 'h = 0' used;

ft wrong value of p

E1B1F

2

(b) (i) $x^2 = -1 - \frac{1}{2} = -1.5$

M1

ft wrong gradient

A1F

2

(ii) Tangent at A drawn

M1

α and x_2 shown correctly

dep't only on the last M1

A1

2

[12]

Q18.

(a) $A(-1, \pi)$

B1

$B\left(0, \frac{\pi}{2}\right)$

B1

2

(b) $\cos^{-1} x - 3x - 1 = 0$

$f(0.1) = 0.17$ allow 0.2, 0.1

Or comparing 'sides'

M1

$f(0.2) = -0.23$ allow -0.2

Change of sign $\therefore$ root

A1
2

(c) $x_1 = 0.1$

M1

$$x_2 = 0.1569 = 0.157$$

A1

$$x_3 = 0.1378 = 0.138$$

$$x_4 = 0.144$$

A1
3

[7]

Q19.

(a) $f(0.5) = -0.875$

M1

$$f(1) = 2$$

Change of sign $\therefore$ root

A1
2

(b) $x^3 + 4x - 3 = 0$

$$4x = 3 - x^3$$

B1
1

$$x = \frac{3 - x^3}{4}$$

AG

(c) (i) $x_1 = 0.5$

M1

$$x_2 = 0.71875 \quad 0.72 \quad \text{AWRT}$$

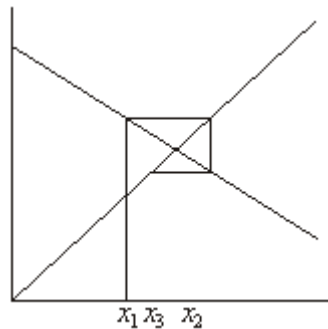
A1

$$x_3 = 0.66$$

A1

3

(ii)



For cobweb, x_1 to curve

M1

For x_2

All correct

A1

A1

3

[9]

Q20.

(a) $f(2) = -1$
 $f(2.1) = +0.161$

both attempted

M1

change of sign $\therefore 2 < \alpha < 2.1$

A1

2

(b) $x^3 - x - 7 = 0$
 $x^3 = x + 7$
 $x = \sqrt[3]{x+7}$

AG

B1

1

(c) $x_1 = 2$

M1

$$x_2 = 2.0801\dots$$

AWRT 2.08

A1

$$x_3 = 2.0862\dots$$

AWRT 2.09

$$x_4 = 2.09$$

A1

3

[6]

Q21.

(a) $f'(x) = 3x^2 - 2$

B1

$$x_2 = 1 - \frac{-2}{1} = 3$$

M1A1

3

(b) Tangent at P drawn

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x_1 and x_2 shown correctly

B1

2

(c) $f(2) = 3 > 0$, so root < 2

E1 for incomplete explanation

E2,1

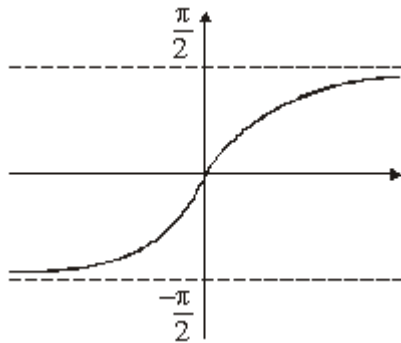
2

(d) $x_2 = 1.6 - \frac{-0.104}{5.68} \approx 1.618$

M1A1

Q22.

(a)



shape

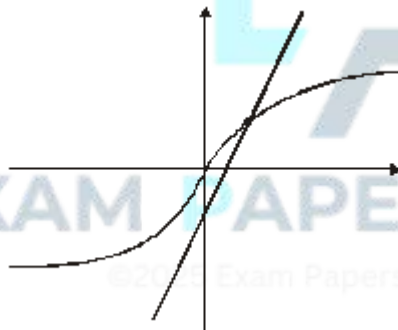
B1

asymptotes (shown or stated) ($\frac{\pi}{2}$ seen)

B1

2

(b) (i)



sketch of $2x - 1$

B1

correct

B1

2

(ii) $\tan^{-1} x - 2x + 1 = 0$

$f(0.8) = 0.07$

M1

$$f(0.9) = -0.07$$

change of sign $\therefore$ root

allow +ve, -ve

A0 if $f(0.8)$, $f(0.9)$ wrong

A1

2

(c) $(x_1 = 0.8)$

attempt at x_2

M1

$$x_2 = 0.837 \text{ (37) } \dots$$

for x_2

A1

$$x_3 = 0.85$$

for x_3

A1

3

[9]



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